Data services in a cloud native system are generally off-the-shelf components like databases and message brokers. There are a few properties you should consider to ensure you choose the most suitable technology.

- *Scalability*—Cloud native applications can scale in and out dynamically. Data services are no different: they should scale to adapt to increasing or decreasing workloads. The new challenge is scaling while ensuring safe access to the data storage. The amount of data flying through a system in the cloud is larger than ever, and there can be sudden increments, so data services should support the likelihood of increasing workloads and be resilient.
- *Resilience*—Much like cloud native applications, data services should be resilient to failures. The new aspect here is that the data persisted using a specific storage technology should also be resilient. One of the key strategies for ensuring your data is resilient and preventing data loss is duplication. Replicating data across different clusters and geographical zones makes it even more resilient, but this comes at a cost. Data services like relational databases allow replication while ensuring data consistency. Others, like some non-relational databases, provide a high level of resilience but can’t always guarantee data consistency (they offer what is referred to as *eventual consistency*).
- *Performance*—The way data is duplicated can affect performance, which is also limited by the I/O access latency of the specific storage technology and the network latency. Where the storage is located compared to the data services relying on it becomes important—this is a concern that we haven’t encountered with cloud native applications.